ECE343 Spring 2025

Objective 9.23:

UPW

Characteristics



SCHEDULE

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
Chapter 6: Steady Current & Conductivity			
28	Current density, Conductivity, Resistance	Ch. 6	
Chapter 7: Magnetostatics			
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2 **GR 2**
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
Chapter 8: Time-Varying Fields			
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
Chapter 9: Plane Waves in Lossless Media			
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

Ch 9 Learning Objectives

9.1 I understand the process of decoupling and finding solutions for Maxwell's equations for dynamic fields.

9.2 I can articulate what a Uniform Plane Wave (UPW) is and describe key properties that relate the electric and magnetic field.

9.3 I can calculate the electric and magnetic fields for a UPW.

9.4 I can articulate the concept of polarization and determine the polarization for various UPWs.

9.5 I can determine the reflection and transmission of UPW at a boundary for Transverse Electric (TE) and Transverse Magnetic (TM) waves, using boundary conditions.

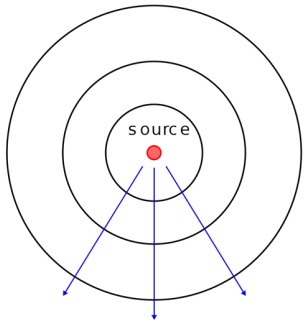
Types of Waves: By Shape of Wave Front

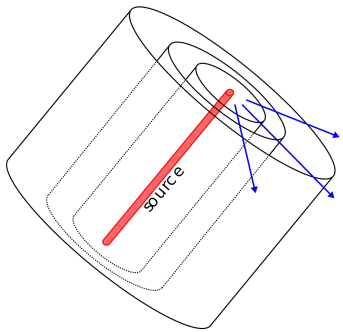
Surfaces of constant phase can be:

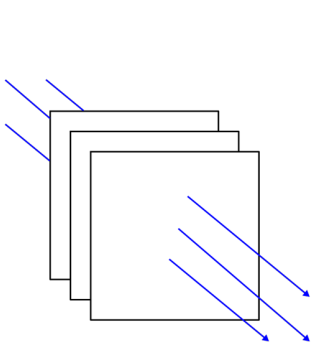
- Spherical
  Appropriate when source is small relative to domain

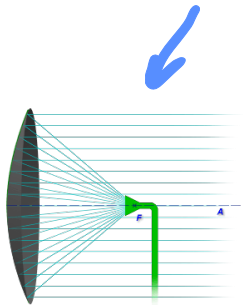
- Cylindrical
  Appropriate when source is shaped like a line.

- Planar
  Appropriate when wave is observed locally or when wave is intentionally guided to be planar.









Uniform Plane Waves

Assumptions:

- Simple Media:
  - Linear
  - Isotropic
  - Homogeneous
  - Non-dispersive
- Magnitude and phase are constant in a plane perpendicular to direction of propagation (assume z)

$\frac{\partial \tilde{E}}{\partial x} = \frac{\partial \tilde{E}}{\partial y} = \frac{\partial \tilde{H}}{\partial x} = \frac{\partial \tilde{H}}{\partial y} = 0$

Which means:

$\frac{\partial}{\partial x_{x,y}} \tilde{E}_x = \frac{\partial}{\partial x_{x,y}} \tilde{E}_y = \frac{\partial}{\partial x_{x,y}} \tilde{E}_z = 0$

$\frac{\partial}{\partial x_{x,y}} \tilde{H}_x = \frac{\partial}{\partial x_{x,y}} \tilde{H}_y = \frac{\partial}{\partial x_{x,y}} \tilde{H}_z = 0$

Now, the wave equation can be written as:

$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right](\hat{x}\tilde{E}_x + \hat{y}\tilde{E}_y + \hat{z}\tilde{E}_z) - \gamma^2(\hat{x}\tilde{E}_x + \hat{y}\tilde{E}_y + \hat{z}\tilde{E}_z) = 0$

Uniform Plane Waves: Fields

Each component separately must equal zero, allowing us to write 3 equations:

So, the UPW conditions reduce these equations to:

$$\frac{\partial^2}{\partial z^2}\tilde{E}_{x,y,z} - \gamma^2 \tilde{E}_{x,y,z} = 0$$

$$\tilde{\mathbf{E}} = \hat{x}\tilde{E}_x(z) + \hat{y}\tilde{E}_y(z)$$

We can make one more simplification, by considering Ampere's Law:

$$\nabla \times \tilde{\mathbf{H}} = (\sigma + j\omega\epsilon)\tilde{\mathbf{E}}$$

$$\hat{z}\cdot(\nabla \times \tilde{\mathbf{H}}) = \hat{z}\cdot[(\sigma + j\omega\epsilon)\tilde{\mathbf{E}}]$$

$$\frac{\partial \tilde{H}_x}{\partial y} - \frac{\partial \tilde{H}_y}{\partial x} = (\sigma + j\omega\epsilon)\tilde{E}_z$$

$$\frac{0 - 0}{\sigma - j\omega\epsilon} = \tilde{E}_z$$

$$0 = \tilde{E}_z$$

$$\frac{\partial^2}{\partial z^2}\tilde{E}_{x,y} - \gamma^2 \tilde{E}_{x,y} = 0$$

↗ 2 equations

Uniform Plane Waves: Fields

$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta$

We know how to find solutions for the 2nd order ODEs (right?!)

$$\frac{\partial^2}{\partial z^2}\tilde{E}_x - \gamma^2 \tilde{E}_x = 0$$

$$\tilde{E}_x(z) = E_{x0}^+ e^{-\gamma z} + E_{x0}^- e^{+\gamma z}$$

$$\tilde{E}_y(z) = E_{y0}^+ e^{-\gamma z} + E_{y0}^- e^{+\gamma z}$$

$$\tilde{\vec{E}} = \hat{x}\tilde{E}_x + \hat{y}\tilde{E}_y$$

$$\tilde{H}_x = H_{x0}^+ e^{-\gamma z} + H_{x0}^- e^{+\gamma z}$$

$$\tilde{H}_y =$$

$$\tilde{\vec{H}} = \hat{x}\tilde{H}_x + \hat{y}\tilde{H}_y$$

Now, back to time domain...multiply by $e^{j\omega t}$ and take the real part.

$$\tilde{E}_{x,y}(z,t) = \mathrm{Re}\left[E_{x0}^+ e^{-\gamma z} e^{j\omega t} + E_{x0}^- e^{+\gamma z} e^{j\omega t}\right]$$

$$e^{-\gamma z} = e^{-(\alpha+j\beta)z} = e^{-\alpha z} e^{j\beta z}$$

$$\mathrm{Re}\left[e^{-\alpha z} e^{-j\beta z} e^{j\omega t}\right] = \mathrm{Re}\left[e^{-\alpha z} e^{j(\omega t - \beta z)}\right]$$

$$e^{-\alpha z}\,\mathrm{Re}\left[\cos(\omega t - \beta z) + j\sin(\omega t - \beta z)\right]$$

$$E_{x0}^+ e^{-\alpha z}\cos(\omega t - \beta z) + E_{x0}^- e^{+\alpha z}\cos(\omega t + \beta z)$$

UPW Summary

Maxwell's Equations

$$\text{Gauss} \longrightarrow \boldsymbol{\nabla} \cdot \tilde{E} = \frac{\rho_v}{\varepsilon}$$

$$\text{Gauss} \longrightarrow \boldsymbol{\nabla} \cdot \tilde{H} = 0$$

$$\text{Ampere} \longrightarrow \boldsymbol{\nabla} \times \tilde{H} = (\sigma + j\omega\varepsilon)\,\tilde{E}$$

$$\text{Faraday} \longrightarrow \boldsymbol{\nabla} \times \tilde{E} = -j\omega\mu\tilde{H}$$

Helmholtz

$$\nabla^2\tilde{E} - \gamma^2\tilde{E} = 0$$

$$\nabla^2\tilde{H} - \gamma^2\tilde{H} = 0$$

$$\gamma^2 = j\omega\mu\,(\sigma + j\omega\varepsilon)$$

Field Solutions

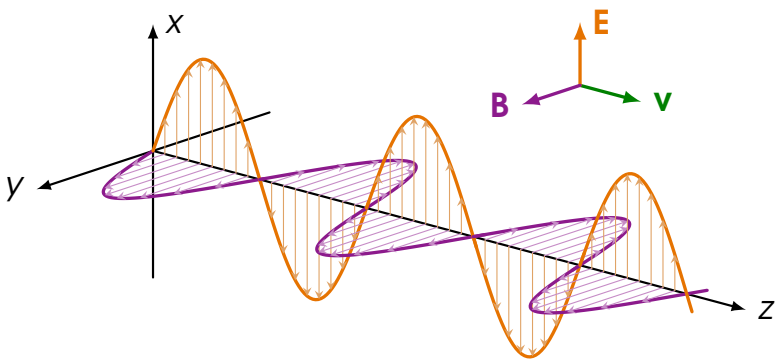
$$E_{x,y}(z,t) = E^{+}_{x0,y0}e^{-\alpha z}\cos(\omega t - \beta z) + E^{-}_{x0,y0}e^{\alpha z}\cos(\omega t + \beta z)$$

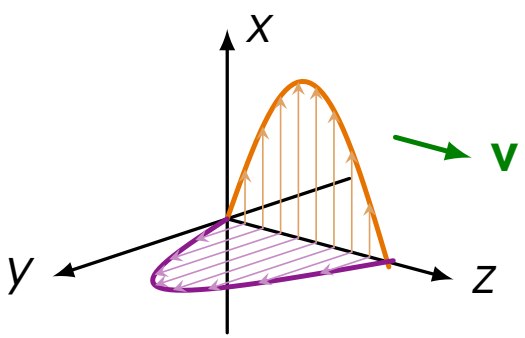
$$H_{x,y}(z,t) = H^{+}_{x0,y0}e^{-\alpha z}\cos(\omega t - \beta z) + H^{-}_{x0,y0}e^{\alpha z}\cos(\omega t + \beta z)$$

Uniform Plane Waves: Interpretation

Observations

- For a UPW, the electric and magnetic field _____ are _____ to propagation.
- The direction of propagation is _____ to the _____ and _____
- The _____ of the _____ and _____ are related





Chapter 3: Lesson 5

$$V(z,t) = \underbrace{V_0^+ e^{-\alpha z}\cos(\omega t - \beta z)}_{\text{Forward Wave}} + \underbrace{V_0^- e^{\alpha z}\cos(\omega t + \beta z)}_{\text{Reverse Wave}}$$

- $V_0^{\pm} \longrightarrow$ Amplitude coefficients
- $e^{\pm\alpha z} \longrightarrow$ Attenuation term
- $\cos(\omega t \pm \beta z) \longrightarrow$ Time and space-varying wave

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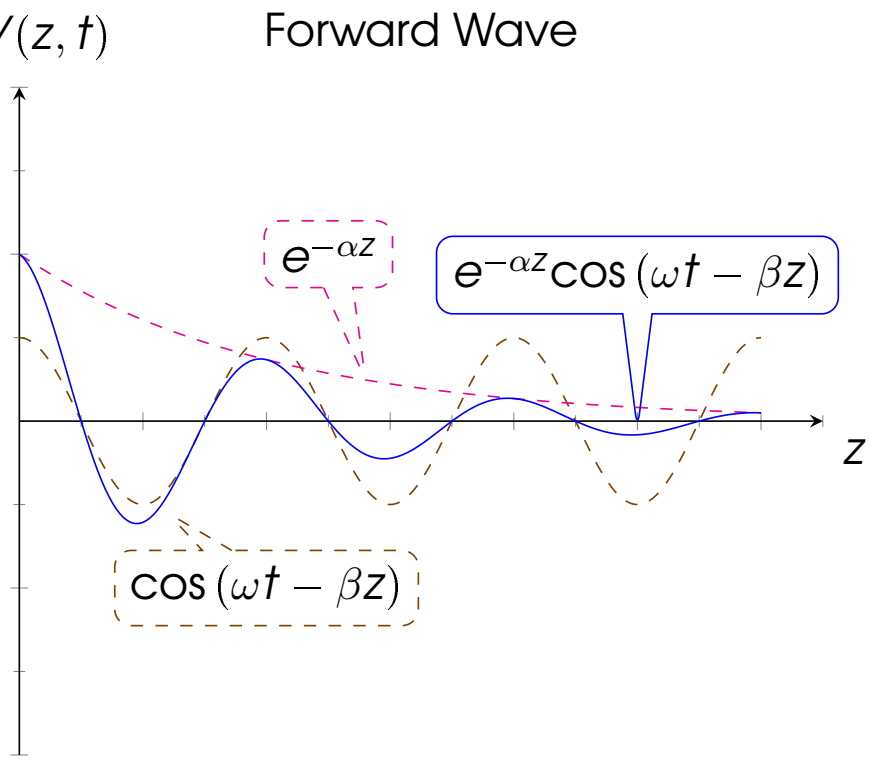
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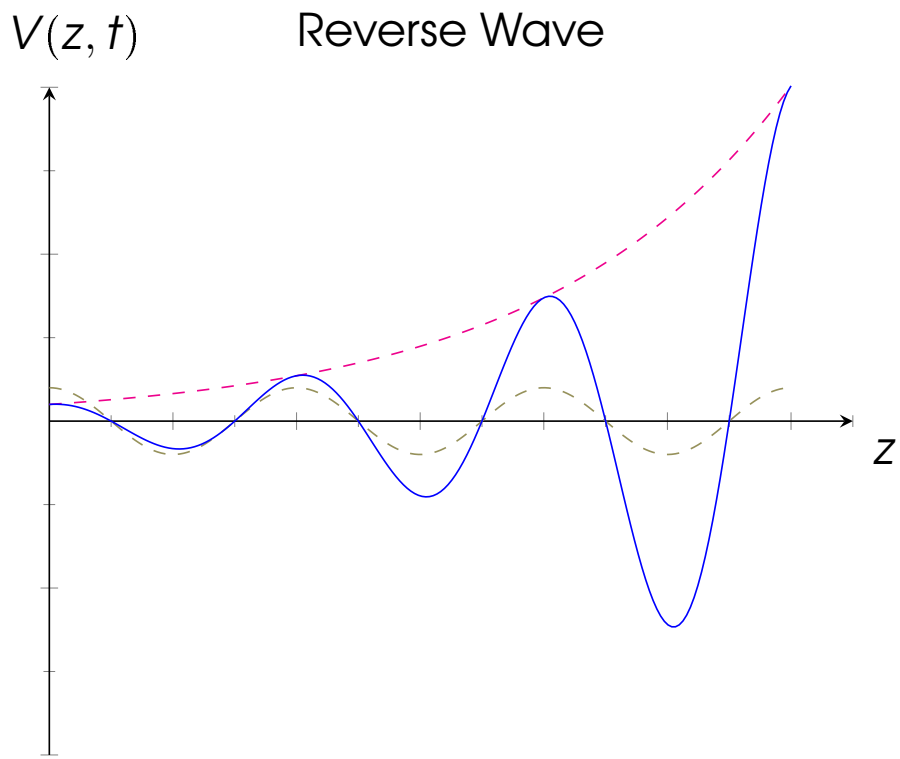
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Uniform Plane Waves: Characteristics

Assume an $\hat{\mathbf{x}}$ Electric field...

$$\hat{x}\tilde{E}_x(z)$$

$$\nabla \times \tilde{E} = -j\omega\mu\tilde{H}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \tilde{E}_x & 0 & 0 \end{vmatrix}$$

$$\hat{x}(0) - \hat{y}\left(0 - \frac{d\tilde{E}_x}{dz}\right) + \hat{z}(0-0) = -j\omega\mu\tilde{H}$$

$$\hat{y}\frac{d\tilde{E}_x}{dz}\left(\frac{1}{-j\omega\mu}\right) = \tilde{H}.$$

$$\hat{y}\left[\frac{d}{dz}\frac{E_{0x}^{+}}{-j\omega\mu}e^{-\gamma z} + \frac{d}{dz}\frac{E_{0x}^{-}}{j\omega\mu}e^{+\gamma z}\right] = \tilde{H}$$

$$\hat{y}\left[\frac{-\gamma}{-j\omega\mu}E_{0x}^{+}e^{-\gamma z} + \frac{\gamma}{-j\omega\mu}E_{0x}^{-}e^{+\gamma z}\right] = \tilde{H}$$

$$\hat{y}\left[H_{0y}^{+}e^{-\gamma z} + H_{0y}^{-}e^{+\gamma z}\right] = \tilde{H}$$

Intrinsic impedance is like characteristic impedance...

$$\eta = \frac{E_{0x}^{+}}{H_{0y}^{+}} = \frac{E_{0x}^{+}}{\frac{\gamma}{j\omega\mu}E_{0x}^{+}} = \frac{j\omega\mu}{\gamma}$$

$$\eta = \frac{j\omega\mu}{\sqrt{j\omega\mu(\sigma + j\omega\epsilon)}} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

$\tilde{E}$, $\tilde{H}$, and propagation ($\hat{a}_p$) are **generally** related:

$$\tilde{E} = -\eta\,\hat{a}_p \times \tilde{H}$$

$$\tilde{H} = \frac{1}{\eta}\hat{a}_p \times \tilde{E}$$

Uniform Plane Waves: General

Summary

For $\hat{z}$-propagating waves!

$$\vec{E}(z,t) = \hat{\mathbf{x}}E_x(z,t) + \hat{\mathbf{y}}E_y(z,t)$$

$$E_{x,y}(z,t) = E^{\pm}_{x0,y0}e^{\mp\alpha z}\cos(\omega t \mp \beta z)$$

$$\tilde{H}(z,t) = \hat{\mathbf{x}}H_x(z,t) + \hat{\mathbf{y}}H_y(z,t)$$

$$H_{x,y}(z,t) = H^{\pm}_{x0,y0}e^{\mp\alpha z}\cos(\omega t \mp \beta z)$$

$$\tilde{E} = -\eta\hat{a}_p \times \tilde{H} \qquad \tilde{H} = \frac{1}{\eta}\hat{a}_p \times \tilde{E} \qquad \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

Uniform Plane Waves: Dielectrics

For dielectrics ($\sigma = 0, \varepsilon = \varepsilon_r\varepsilon_0, \mu = \mu_0$):

$$\gamma = \sqrt{j\omega\mu_0(0 + j\omega\varepsilon_r\varepsilon_0)} = \sqrt{-\omega^2\mu_0\varepsilon_r\varepsilon_0}$$

$$\gamma = j\omega\sqrt{\mu_0\varepsilon_0}\sqrt{\varepsilon_r} = \alpha + j\beta$$

$$\alpha = 0 \qquad \beta = \omega\sqrt{\mu_0\varepsilon_0}\sqrt{\varepsilon_r}$$

$$\eta = \sqrt{\frac{j\omega\mu_0}{0 + j\omega\varepsilon_r\varepsilon_0}} = \sqrt{\frac{\mu_0}{\varepsilon_0}}\sqrt{\frac{1}{\varepsilon_r}}$$

$$= \frac{120\pi\,\Omega}{\sqrt{\varepsilon_r}} \approx \frac{377\,\Omega}{\sqrt{\varepsilon_r}}$$

$$u_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{\mu_0\varepsilon_0}\sqrt{\varepsilon_r}} = \frac{c}{\sqrt{\varepsilon_r}}$$

Free Space

$\eta = 120\pi\,\Omega = 377\,\Omega$

$u_p = c$

Silicon, $\varepsilon_r = 11.6$

$\eta = 110.7\,\Omega$

$u_p = 0.3c$

Uniform Plane Waves: Conductors

For conductors: $\sigma \gg \omega\epsilon$, $\frac{\sigma}{\omega\epsilon} \gg 1$

$$\gamma = \sqrt{j\omega\mu_0(\sigma + j\omega\epsilon)}$$

$$= \sqrt{j\omega\mu_0(\sigma)}$$

$$= \sqrt{j}\sqrt{\omega\mu_0\sigma}$$

$$= (e^{j\pi/2})^{1/2}\sqrt{\omega\mu_0\sigma}$$

$$e^{j\pi/4}\sqrt{\omega\mu_0\sigma} = \sqrt{\omega\mu_0\sigma}\left(\cos(\pi/4) + j\sin(\pi/4)\right)$$

$$\alpha + j\beta$$

$$\alpha = \beta = \sqrt{\omega\mu_0\sigma}\,\frac{\sqrt{2}}{2}$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\eta = \sqrt{\frac{j\omega\mu_0}{\sigma + j\omega\epsilon_0}}$$

$$\eta = \sqrt{\frac{j\omega\mu_0}{\sigma}}$$

$$\eta = e^{j\pi/4}\sqrt{\frac{\omega\mu_0}{\sigma}}$$

$$\tilde{E} = -\eta\hat{a}_p \times \tilde{H}$$

$$\tilde{E} = \eta e^{j\pi/4}\hat{a} \times \tilde{H}$$

What does this mean?!?!

Uniform Plane Waves: Conductors

$\tilde{E} \overset{?}{\leftrightarrow} \tilde{H}$

$E_0^+ \approx e^{j\pi/4} \sqrt{\frac{\omega\mu_0}{\sigma}}\, H_0^+$

Dispersion

$v_p = \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\frac{\omega\mu_0\sigma}{2}}}$

$v_p = \sqrt{\frac{2\omega}{\mu_0\sigma}}$

Skin Depth $\sim e^{-\alpha z}$

Define δ as skin depth, where $\tilde{E}(z = \delta) = E_0 e^{-1}$:

(or) $\downarrow$ $\quad$ $\swarrow$ $0.37\, E_0$

$\alpha z = 1$

$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu_0\sigma}}$